# SnapShield AI

## On-Device Privacy Redaction for Snapdragon-Powered Windows PCs

**Challenge:** Snapdragon AI Lab Build & Present Challenge - Qualcomm

**Participant:** Vinay Kumar

**One-line pitch:** SnapShield AI keeps sensitive images on-device, detects privacy-sensitive regions locally, and redacts them before sharing - with a Snapdragon NPU/QNN execution path designed into the runtime.

## 1. Problem

People frequently share screenshots, photos, meeting captures, classroom images, and device photos that contain faces or other sensitive visual information. Cloud-based processing can add latency, dependency, and privacy concerns. Users need a fast, simple way to sanitize an image before it leaves their machine.

## 2. Proposed Solution

SnapShield AI is a lightweight local application that accepts an image, detects privacy-sensitive content, applies redaction, and lets the user inspect and export the result. The prototype combines a YOLOv5n object detector in ONNX format with local face detection. A user selects a privacy policy such as People + Faces, People + Devices, or Maximum Privacy.

## 3. Snapdragon Optimization

- ONNX Runtime is used as the inference abstraction so the same model pipeline can be deployed across development and Snapdragon environments.
- When the Qualcomm QNN Execution Provider is available, the application selects it before the CPU execution provider.
- QNN supports Windows devices with Snapdragon SoCs and its HTP backend is the NPU-oriented execution target.
- The application keeps a CPU fallback so development and testing can proceed on non-Snapdragon hardware without changing the product workflow.
- Runtime telemetry is shown in the UI: execution provider, target backend, model availability, detection count, and measured request latency.

## 4. User Flow

1. Select an image.

2. Choose a privacy policy.
3. Run local detection and redaction.
4. Inspect original and redacted views side-by-side.
5. Download the redacted image.

## 5. Technical Architecture

- Frontend: HTML/CSS/JavaScript local browser interface.
- Backend: Python + Flask local API.
- Vision: YOLOv5n ONNX + OpenCV face detection and redaction.
- Runtime: ONNX Runtime with QNN-first provider selection on Snapdragon.
- Deployment target: Windows 11 on ARM64 Snapdragon PCs; portable CPU development path.

## 6. Innovation & Use Case

The project turns on-device AI acceleration into a practical privacy workflow rather than a model benchmark alone. The user can see what the system detected, what was redacted, and which execution provider is configured, making the hardware-aware behavior visible and understandable.

## 7. Current Prototype Scope

| Area | Status / plan |
|---|---|
| Available now | Local Flask UI, image upload, policy selection, face redaction, ONNX detector integration, runtime telemetry, download flow. |
| Snapdragon path | QNN Execution Provider preferred when installed on a Windows on Snapdragon environment. |
| Next iteration | Benchmark on physical Snapdragon hardware, add document/QR/OCR redaction, and package as a Windows desktop executable. |

## 8. Performance Evidence and Claim Boundary

Qualcomm AI Hub currently publishes a YOLOv5 profile targeting a Snapdragon X Elite CRD on Windows 11. That profile lists an estimated 14.8 ms inference time, 47 MB peak memory, and NPU compute. These are Qualcomm AI Hub reference measurements for the published profile and are not claimed as measurements produced by this prototype. A final Snapdragon hardware benchmark should be added before making device-specific performance claims.

## 9. Submission Relevance

The challenge requires the proposed solution to be designed, developed, or intended to be optimized for Snapdragon-powered HP PCs. SnapShield AI is architected around a Qualcomm-supported QNN

execution path on Windows on Snapdragon while retaining a portable CPU development mode. The challenge listing also states that the participant should own a Snapdragon-powered laptop; that eligibility field should therefore be answered only according to the participant's actual hardware ownership.

## 10. References


- **Qualcomm - Snapdragon AI Lab -** https://www.qualcomm.com/snapdragon/ai-lab
- **Qualcomm - Snapdragon AI Hub / developer resources -** https://aihub.qualcomm.com/
- **Qualcomm AI Hub - YOLOv5 Snapdragon X Elite profile -** https://aihub.qualcomm.com/jobs/jpxl31x1p
- **ONNX Runtime - QNN Execution Provider -** https://onnxruntime.ai/docs/execution-providers/QNN-ExecutionProvider.html
- **Challenge listing - Snapdragon AI Lab Build & Present Challenge -** https://unstop.com/competitions/1748893